LLM-based Code-Switched Text Generation for Grammatical Error Correction

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Abstract

With the rise of globalisation, code-switching (CSW) has become a ubiquitous part of multilingual conversation, posing new challenges for natural language processing (NLP), especially in Grammatical Error Correction (GEC). This work explores the complexities of applying GEC systems to CSW texts. Our objectives include evaluating the performance of state-of-the-art GEC systems on an authentic CSW dataset from English as a Second Language (ESL) learners, exploring synthetic data generation as a solution to data scarcity, and developing a model capable of correcting grammatical errors in monolingual and CSW texts. We generated synthetic CSW GEC data, resulting in one of the first substantial datasets for this task, and showed that a model trained on this data is capable of significant improvements over existing systems. This work targets ESL learners, aiming to provide educational technologies that aid in the development of their English grammatical correctness without constraining their natural multilingualism.

1 Introduction

Code-switching (CSW), the practice of fluidly alternating between two or more languages in conversation, has become commonplace in recent years. This linguistic phenomenon, emerging as a natural consequence of multilingualism, is now widely accepted in social and professional settings (Yow et al., 2018). Many works have highlighted the utility and cultural importance of CSW in general conversation (Beatty-Martínez et al., 2020; Falbo and LaCroix, 2021). Further research indicates that these advantages extend to language learning, with CSW offering many pedagogical benefits. These include increasing students' access to content and improving their confidence. Nguyen et al. (2022) discuss the mechanisms for this, where students use a familiar language to grasp foreign, complex concepts. CSW can also serve as a scaffolding tool, helping to bridge gaps in a student's comprehension of a language and enabling them to build upon existing knowledge. These benefits reduce the barriers between a student and their target language and help promote a learning environment conducive with active exploration and deeper understanding. Therefore, it is essential that English as a Second Language (ESL) learners are not penalised for expressing their cultural identity through CSW. Grammatical error correction (GEC) is the task of automatically detecting and correcting errors in text. Research on GEC for CSW text remains untouched, yet crucial to ensure language technology is inclusive and reflective of real-world linguistic practices. Figure 1 shows two examples of CSW from our target population with their grammatical corrections.[1]

Example 1: According to the test, [lacks in me → my shortcomings] are 靴下 and ご主人様.
Example 2: When we [call → say] ダッシュボード, do we actually mean a glove compartment in English?

Figure 1: Examples of GEC in ESL learner language.

Despite significant advancements in GEC in recent years, a gap persists in addressing CSW texts, with monolingual GEC datasets labelling CSW as a type of error (Bryant et al., 2019a). There are several reasons for this, the most prominent being the scarcity of high-quality training data, a problem that plagues monolingual GEC systems. The unique linguistic features of CSW, including its variable syntax, semantics and pragmatics, add additional complexity to this task. Monolingual seq2seq GEC models, e.g. T5 (Rothe et al., 2021), struggle with CSW text as they fail to represent the non-English inputs, resulting in their inability to output the CSW text. On the other hand, multilingual seq2seq models and edit-based GEC mod-

[1]The definition of CSW is a subject of ongoing debate. Throughout this work, we use the term CSW to refer specifically to the type of language mixing exhibited by ESL learners.

els like GECToR (Omelianchuk et al., 2020) can handle CSW text but struggle with the ambiguity present at language switching points. This ambiguity challenges the models' ability to accurately correct the text.

This paper aims to bridge this gap. Firstly, to address the data scarcity issue, we propose a method for generating high-quality synthetic CSW GEC data, using which we produce, to our knowledge, one of the first substantial datasets labelled for this task[2]. Secondly, we train a token classification-style GEC system, tailored to correct errors in texts produced by ESL learners. This demographic is significant for our study as they not only present consistent CSW patterns but also stand to benefit greatly from a GEC system capable of handling CSW text.

2 Data

2.1 Genuine CSW GEC Dataset

One of the only datasets labelled for GEC which does not remove CSW text, is the Lang-8 dataset (Mizumoto et al., 2013), sourced from the Lang-8 language learning platform. This dataset, when filtered to contain entries where CSW is present, offers a foundation of authentic data, comprises 5,875 pairs of ungrammatical and corrected sentences across 6 CSW language pairs: English-Japanese (81.9%), English-Korean (13.0%), English-Traditional Chinese (3.4%), English-Russian (1.2%), English-Thai (0.5%) and English-Arabic (0.1%).

The crowd-sourced nature of Lang-8 required manual validation to ensure accuracy. We tasked an annotator with the responsibility of verifying the original corrections in the dataset, as well as combing for missed errors, incorrect annotations and over-annotations.

2.2 Synthetic CSW GEC Data Generation

Given the small size of the available CSW GEC dataset, we introduced a 2-step approach to synthetic CSW GEC data generation. First, we generated grammatically correct CSW sentences. This is followed by the introduction of errors.

2.2.1 Step 1: CSW Text Generation

Three different synthetic data generation techniques have been explored to generate CSW data.

Translation-based CSW Text Generation required a monolingual corpus, a machine translation (MT) algorithm, and a sentence parser. To generate a CSW utterance, we used the Stanford Parser v4.5.4[3] (Manning et al., 2014) to build a syntactic parse tree. We then randomly selected and translated a subtree using the ArgosTranslate MT package[4] (Finlay, 2023; Klein et al., 2017). This method generates plausible CSW text. However, performance is dependent on the strength of the parsing and translation algorithms; and the style of language within the corpus. To approximate the style of our authentic CSW text, we used corrected monolingual sentences from the Lang-8 corpus.

Parallel Corpus-based CSW Text Generation avoids the need for a translation algorithm. Instead, we used the same Stanford Parser, this time with Spanish, French and German configurations; and the AWESOME word-alignment model[5] (Dou and Neubig, 2021), to identify parts of parallel corpora labelled for MT with similar syntactic structure. For this method, we use the Europarl corpus (Koehn et al., 2012) due to the grammatical quality of its English component. Under the Equivalence Constraint Theory (Rizvi et al., 2021), these areas are where CSW is likely to take place. We, therefore, randomly chose overlapping subtrees as candidates for injection of non-English text. Although this method does not require MT, it is reliant on performant word-alignment and parsing systems; these are rare for many languages.

LLM Prompting-based CSW Text Generation The other methods of generating CSW text rely on injecting a second language into existing monolingual corpora. Hence, they are not able to recreate one of the main switching styles shown by ESL learners - CSW as a genuine pragmatic strategy. A common reason for this style of switching is when quoting another language. It is difficult to recreate this style using a monolingual foundation as sentences like *The Japanese word for "dog" is "犬"* seldom appear in authentic monolingual corpora.

To generate diverse CSW texts without relying on existing corpora or inaccurate alignment algorithms, we leveraged the strong general knowledge of Large Language Models (LLMs). We demonstrated that OpenAI's GPT-3.5 (Brown et al., 2020) can create high-quality CSW sentences when

[2]This dataset will be available upon publication.

[3]This can be downloaded from the CoreNLP website.

[4]We used ArgosTranslate v1.8.0, available on GitHub.

[5]The authors shared their model on HuggingFace.

Metric	Genuine CSW	LLM CSW	Translation CSW	Corpus CSW
CMI	15.52	16.14	27.81	11.42
M-Index	0.007	0.004	0.015	0.006
I-Index	0.21	0.21	0.30	0.20
Burstiness	-0.07	-0.04	0.03	-0.11
CF1	6.38	5.82	17.13	2.55
CF2	19.77	19.03	31.11	16.04
CF3	18.34	17.61	30.05	14.20

Table 1: Quantitative Description of the Genuine and Generated CSW Datasets Using Various CSW Metrics.

shown examples of authentic utterances. Along with genuine CSW texts, we supplied a one-shot example of how to use the switching styles of an existing CSW text to generate a new sentence.[6]

Comparison of Synthetic CSW Text We used several CSW metrics to quantify the qualities of CSW texts: Code Mixing Index (CMI) (Gambäck and Das, 2016), Multilingual Index (M-Index) (Barnett et al., 2000), Probability of Switching (I-Index) (Guzmán et al., 2017), Burstiness (Goh and Barabási, 2008), and Complexity Factor (CF1-3) (Ghosh et al., 2017). Table 1 shows the value of each metric for our genuine CSW dataset, as well as for these 3 synthetic CSW datasets. We can see that the LLM prompting-based dataset was superior in its similarity to the authentic CSW data. Using this method, we generated a corpus of 73,293 utterances covering over 20 language pairs, including the 6 language pairs in the original dataset.[7]

2.2.2 Step 2: Synthetic Error Generation

Several works have shown the effectiveness of rule-based error injection for GEC data generation. Many use the PIE-synthetic dataset (Awasthi et al., 2019), a perturbed version of the 1BW corpus (Chelba et al., 2013). For each sentence, the authors introduce between 0 and 4 errors of random type. We extended this work by introducing a new subset of error types that are not only more common in ESL learners, but also are areas where the SOTA performance collapses when faced with CSW text: noun, pronoun, word order, determiner, and punctuation errors.[8]

To increase the diversity of errors, we adopted a second style of error injection, Backtranslation (Stahlberg and Kumar, 2021). By swapping the source and target sentences of a monolingual dataset, we trained a GECToR-based system to induce errors in our synthetic CSW sentences.

Using these methods, we created two datasets: Syn-CSW PIE and Syn-CSW Rev-GECToR. After removing pairs containing no injected errors, we are left with 70,181 and 18,160 sentences each.

3 CSW GEC Systems

For our GEC system targeting CSW texts, we chose a GECToR model (Omelianchuk et al., 2020), with a RoBERTa-base foundation, due to its proven efficacy with limited training data and stronger performance on CSW texts compared to seq2seq models. We added a new CSW class to the error detection head, adding the ability to detect CSW tokens.

Following Tarnavskyi et al. (2022), we used a 3-stage training schedule. In the first, we used the same distilled 1BW corpus, and added all our synthetic CSW GEC data. For the second, we used several GEC datasets: NUCLE (Dahlmeier et al., 2023), FCE (Yannakoudakis et al., 2011), W&I Locness (Bryant et al., 2019b), Lang-8 and our 2 synthetic CSW datasets. As our genuine CSW dataset is a subset of the Lang-8 corpus, we checked and removed any duplicates. Following previous works, we finished training using the W&I Locness dataset due to its superior quality. In this final stage, we added a sampled subset of our synthetic CSW sentences and 90% of our authentic CSW data, ensuring exposure to synthetic and genuine CSW text. At each stage, we reserved 5% of the data for validation.[9] Finally, we tuned inference parameters using a grid-search to optimise the $F_{0.5}$ on the final validation set.

4 Results and Analysis

4.1 Baseline Comparisons

We compared our model against two well-established systems: a RoBERTa-base GECToR

[6]The full prompt can be seen in Appendix A.

[7]The LLM does not always generate the language pairs we ask for. However, these sentences are still included in the dataset categorised under their actual language pair.

[8]Error type analysis is presented in Appendix B.

[9]An exact breakdown of contributions by each dataset is given in Appendix C.

Model	BEA-2019 Test			Genuine CSW		
	P	R	$F_{0.5}$	P	R	$F_{0.5}$
Existing GEC systems						
GECToR	77.88	53.07	71.22	71.14	27.08	53.67
T5-Small	62.03	47.19	58.34	11.70	24.98	13.09
Our CSW GEC systems						
Stage 1	67.23	53.88	64.05	66.15	26.04	50.57
Stage 2	72.64	51.73	67.20	65.41	29.93	52.87
Stage 3	74.32	53.40	68.92	84.66	22.92	55.02
Inference Tweaks	69.01	58.40	66.59	76.02	38.67	63.71

Table 2: ERRANT-based Precision, Recall and $F_{0.5}$ Scores of Baselines and Our Model Throughout Training

model (Omelianchuk et al., 2020), with near SOTA performance on the BEA-2019 test set (Bryant et al., 2019b), and a seq2seq T5 model (Rothe et al., 2021). To assess these models, we evaluated their performance on the BEA-2019 test set and the remaining 10% of our authentic CSW data. The ERRANT (Bryant et al., 2017) GEC evaluation results, as outlined in Table 2, demonstrate a clear degradation in performance when these two systems are applied to CSW texts.

4.2 Detailed Model Performance

The progression of our model throughout training provided insights into its evolving capabilities and effectiveness of our synthetic data. We monitored several metrics, including the ERRANT precision, recall and $F_{0.5}$ score, for the BEA-2019 test set and the remaining unused 10% of our genuine CSW dataset. These metrics, as displayed in Table 2, indicate a steady improvement in the ability to handle CSW texts. Notably, the performance on the CSW dataset shows a significant leap in the final stages, where the contribution of our synthetic dataset is largest. This improvement in CSW text handling did slightly compromise the model's performance on monolingual GEC tasks, as seen on the BEA-2019 test set. This suggests a trade-off inherent in specialising the model for CSW contexts. However, our model remains competitive amongst SOTA monolingual GEC systems of its size.

4.3 Inference Tweaking and Error Thresholds

The inference tweaking phase was crucial in tuning the balance between precision and recall. The changes made here, particularly lowering the minimum error thresholds before the model makes an edit, indicated a clear attempt to force the model to make more corrections.[10] While this slightly lowered precision on monolingual errors, it significantly enhanced the performance on CSW text.

4.4 Synthetic Data Impact

The synthetic CSW text and error injection methods were central to this project. The resemblance of our synthetic text to real ESL learner data, as shown by the similarity metrics in Table 1, is a testament to the effectiveness of our chosen generation method. The improvements in $F_{0.5}$ scores provide further evidence of this.

Our extended PIE-synthetic dataset aimed to introduce four error types common in ESL students: noun, pronoun, punctuation and word errors. When compared to the monolingual GECToR, our model is stronger in all of these areas.[11] This provides strong evidence that the targeted approach to error injection was successful in boosting the model's ability in these areas.

5 Conclusion

The primary aim of this paper was to build a GEC system capable of effectively correcting English errors in CSW text, whilst maintaining competitive performance on monolingual data. To address the scarcity of CSW data, we explored methods of generating synthetic CSW text. We used several CSW metrics to establish that the LLM prompting-based approach was the most capable of generating text resembling the content in our genuine dataset. From there, we used two error injection methods to create the first substantial datasets labelled for CSW GEC. This significantly expanded the training data available. Importantly, it also opened up opportunities for future research in CSW GEC and CSW NLP more generally. We demonstrated the efficacy of our synthetic data generation techniques by training the first GEC model aimed at correcting errors in CSW texts. Our model showed a clear improvement in performance on CSW data, surpassing the SOTA in this area.

[10] Implementation details are presented in Appendix D.

[11] Error type analysis of our model is given in Appendix E.

6 Limitations

This research, while comprehensive, encounters several limitations that highlight areas for potential improvement. One primary limitation lies in the overrepresentation of Japanese in the genuine CSW dataset. This raises questions about the model's applicability to a broader range of language pairs. This is an unfortunate consequence of using a dataset sourced from Lang-8, a Japanese language learning network. Although our method demonstrated that it could generate texts from a wider range of language pairs, it is possible that all CSW data used shows a bias towards Japanese styles of CSW. Such a bias in our system could inadvertently lead to reduced accessibility and effectiveness for ESL learners who CSW with languages other than Japanese. If this system were to be used as an aid in ESL education, steps should be taken to ensure that it does not contribute to existing inequalities present in language learning platforms. English learning tools should be accessible regardless of the student's native language and future work should focus on developing more inclusive datasets to help mitigate these risks.

Another possible limitation relates to the style of model chosen. The sequence tagging method was selected due to its lower data requirements, but this decision may have constrained the capabilities of the model. Many of the errors typical of ESL students require complex restructuring of the sentence - a notably difficult task for edit-based GEC systems. Although the data needs are more substantial, it is likely that NMT GEC systems may fare better as they are not constrained by a limited vocabulary of edits.

To assess the likeness of our generated CSW text, we introduced several common CSW metrics. Although useful, these metrics are not very sophisticated, and often struggle to accurately capture the nuances of CSW patterns across different subpopulations. These language patterns can have a substantial impact on the optimal approaches to problems across CSW NLP, and hence, the field would benefit from further research in this area.

Finally, we reported results for a RoBERTa-base GECToR system. Although we also tested other base models, including BERT, DeBERTa and ELECTRA, we did not look at larger models or ensemble systems. Future extensions could explore this area, building upon the observation that larger models or simple voting ensembles can yield better results than the smaller base models (Tarnavskyi et al., 2022).

In summary, whilst the current work makes significant contributions to the field of GEC for CSW text, these limitations indicate crucial areas for further research and development.

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A Example LLM Prompt

Figure 2 shows an example LLM prompt used to generate synthetic CSW sentences from genuine examples. As we are using a private subset of the Lang-8 dataset, we are not permitted to share any of the CSW texts.

B Error Type Analysis of SOTA

Table 3 shows a breakdown of the performance of a single RoBERTa Large-based GECToR system trained purely on monolingual GEC data when applied to two datasets, our genuine CSW dataset and the BEA-2019 (Bryant et al., 2019b) test set. These datasets are approximately the same size. The model used represents a current near-SOTA single model sequence tagging-based GEC system measured using $F_{0.5}$ on the BEA-2019 test set. For brevity, we have removed categories with a low number of examples in either dataset or where performance is not significantly different.

C Training Data Schedule

In this section, we explicitly detail the data used at each stage of the training process.

Stage 1 For the initial pre-training stage, we used the distilled dataset proposed by the SOTA (Tarnavskyi et al., 2022). This dataset was constructed by extracting corrections from the monolingual 1BW corpus (Chelba et al., 2013) using the highest performing GECToR ensemble. Through this dataset, we shuffled our PIE-synthetic CSW dataset. We deemed this dataset to be of lower quality than its Rev-GECToR counterpart. Consequently, it was used earlier in the training process. This provided roughly 1,200,000 examples for the initial training phase of which we split between train and validation sets according to a ratio of 19:1. Our synthetic CSW sentences comprised approximately 5.65% of this dataset. We aimed to keep this percentage small in this phase of the training process to allow the model to first learn to correct errors in monolingual texts. In later stages, we boosted the contribution of the CSW data.

Stage 2 For the second stage, we shuffled several GEC datasets. These are NUCLE (Dahlmeier et al., 2023), FCE (Yannakoudakis et al., 2011), W&I Locness (Bryant et al., 2019b), Lang-8 (Mizumoto et al., 2013) and our 2 newly created CSW datasets. As our genuine CSW dataset is a subset of the private Lang-8 corpus, we checked and removed any duplicates. Table 4 shows the overall contributions of each corpus towards the stage 2 dataset. Similar to the previous stage, the data was split into train and validation sets.

Stage 3 For the final stage, we combined the high quality W&I Locness dataset with a sampled subset of the genuine CSW data and a sampled subset of the synthetic CSW texts. Again, the stage 3 dataset is split into train and validation sets. The remaining unused subset of the genuine CSW dataset was retained for testing purposes. Table 5 details the contributions to this stage from each dataset.

D Inference Hyperparameters

After training our model, we used the validation dataset from stage 3 to tune 2 inference parameters. These are:

- additional_confidence — This value is added to the probability of the *$KEEP* token. If this value is high, recall is likely to decrease and precision increase. The grid search found the best value of this to be 0.
- min_error_probability — For a change to be made to a sentence, the probability of at least one token in the sentence being an error must be higher than the min_error_probability. If this value is high, then precision is likely to be higher and recall lower. The grid search found the best value of this to be 0.4.

E Error Type Analysis of Proposed Model

By exploring the ERRANT error classifications of our proposed model when applied to the CSW test dataset, we can further explore the effectiveness of our synthetic data in addressing the problematic areas identified in Appendix B. A breakdown of the precision, recall and $F_{0.5}$ score for each of the previously identified categories is shown in Table 6.

Settings: [no prose]
For each of the following code-switched sentences, generate a new sentence that uses the same two languages and a similar style of code-switching. The topic should be different. Ensure you use the correct grammar in the English portion of the sentence. Make sure that each sentence contains 2 languages. Only return the sentences and their number. You must follow all of the instructions.

For example, given the source sentence and label:
1. This food is called “ラーメン”.
An acceptable answer would be:
1. This animal is called a “犬”.

Do not include any other information in the generated sentences. The 10 real examples are as follows:

1. [CSW SENTENCE]
2. [CSW SENTENCE]
3. [CSW SENTENCE]
4. [CSW SENTENCE]
5. [CSW SENTENCE]
6. [CSW SENTENCE]
7. [CSW SENTENCE]
8. [CSW SENTENCE]
9. [CSW SENTENCE]
10.[CSW SENTENCE]

Figure 2: An Example LLM Prompt Used to Generate CSW Text

Category	**BEA-2019 Test**				**Genuine CSW**				**$F_{0.5}$ Decrease**
	$F_{0.5}$	**TP**	**FP**	**FN**	**$F_{0.5}$**	**TP**	**FP**	**FN**	
DET	80.45	432	80	205	46.27	472	351	1336	34.18
NOUN	47.85	29	16	94	4.34	21	147	1725	43.51
ORTH	75.96	201	30	198	36.45	181	264	522	39.51
OTHER	39.51	113	77	557	3.55	39	241	4333	35.96
PREP	75.44	263	58	196	39.14	241	251	870	36.30
PRON	66.38	62	19	81	20.71	21	32	274	45.67
PUNCT	80.93	786	165	266	0.35	1	286	284	80.58
VERB	52.59	61	27	167	18.08	49	77	802	34.51
VERB:FORM	81.62	151	30	50	37.20	61	90	155	44.42
VERB:SVA	88.64	128	14	26	57.58	114	79	104	31.06
VERB:TENSE	65.55	145	62	133	33.80	116	172	448	31.75
WO	58.08	23	5	63	6.33	2	11	104	51.75

Table 3: $F_{0.5}$ Scores, TP, FP, FN, and Differences in $F_{0.5}$ Scores (BEA - CSW) for Different Categories in the BEA-2019 Test Split and our Genuine CSW Dataset.

Dataset	Sentences
Lang-8	985,683 *(80.54%)*
W&I Locness	68,608 *(5.61%)*
NUCLE	54,258 *(4.43%)*
FCE	26,929 *(2.20%)*
Syn-CSW PIE	70,181 *(5.73%)*
Syn-CSW Rev-GECToR	18,160 *(1.48%)*
Total	1,223,819

Table 4: Sentence Count and Contribution of Stage 2 Datasets

Dataset	Sentences
W&I Locness	68,608 *(67.23%)*
Syn-CSW Rev-GECToR	18,160 *(17.80%)*
Syn-CSW PIE	10,000 *(9.80%)*
CSW Genuine	5,279 *(5.17%)*
Total	102,047

Table 5: Sentence Count and Contribution of Stage 3 Datasets

Category	P	R	$F_{0.5}$
NOUN	0.2857	0.0833	0.1923
PRON	0.7647	0.4643	0.6771
PUNCT	0.7143	0.1139	0.3460
WO	0.7778	0.2000	0.4930

Table 6: Precision (P), Recall (R) and $F_{0.5}$ Score of Our Proposed Model for Targeted Error Types in the Genuine CSW Test Dataset